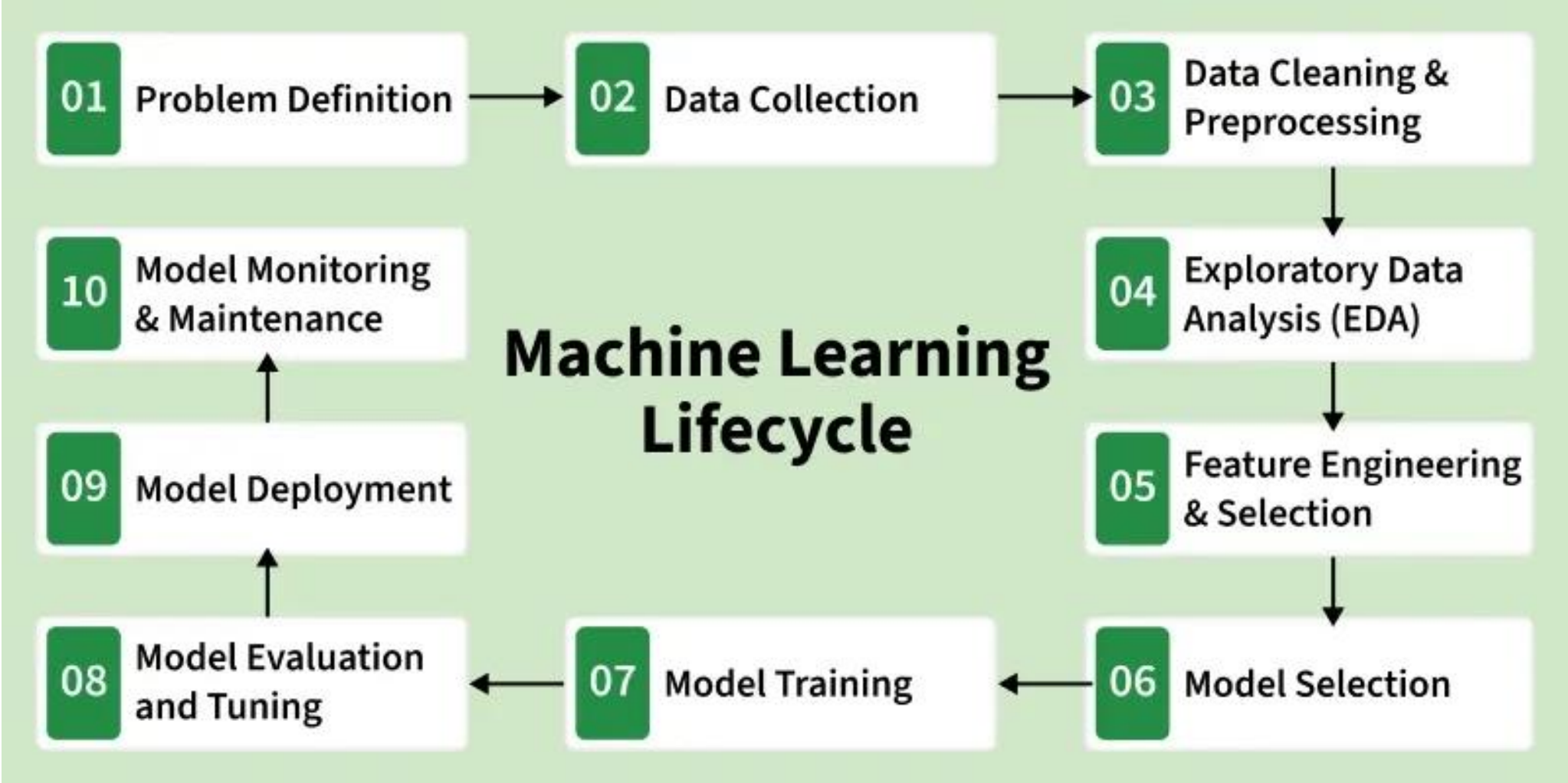
Machine Learning Lifecycle

Machine Learning DSA-447/DGE-301

Machine Learning Pipleline

- Machine Learning Lifecycle is a structured process that defines how ML models are developed, deployed and maintained. It consists of a series of steps that ensure the model is accurate, reliable and scalable.



Machine Learning Pipleline

- **Step 1: Problem Definition**
  - The first step is identifying and clearly defining the business problem.
  - Important things like project objectives, desired outcomes and the scope of the task are carefully designed during this stage.
    1. Collaborate with stakeholders to understand business goals
    2. Define project objectives, scope and success criteria
    3. Ensure clarity in desired outcomes
- **Step 2: Data Collection**
  - Data Collection phase involves systematic collection of datasets that can be used as raw data to train model.
  - The quality and variety of data directly affect the model's performance.
    - Relevance, Quality, Quantity, Diversity

Machine Learning Pipleline

Retail Store:
Predicting Product Demand

Business Problem

Predict the demand for each product

Stakeholder Collaboration

- Store managers explain past inventory issues
- Finance team sets budget limits
- Supply chain staff share delivery time constraints

Problem Definition

- Objective: Forecast weekly product demand
- Scope: Top 50 selling items across 10 branches

Healthcare:
Predicting Patient No-Shows

Business Problem

Predict which patients are likely to miss their appointment

Stakeholder Collaboration

- Doctors explain how no-shows disrupt schedules
- Reception staff describe booking issues
- Admin team highlights revenue loss

Problem Definition

- Objective: Identify high-risk no-show patients
- Scope: Use past appointment data only
- Success Criteria: At least 30% accuracy, reduce no-shows by 25

Education:
Predicting Students Who Need Support

Business Problem

Predict which students are likely to struggle in a course

Stakeholder Collaboration

- Instructors share performance patterns
- Academic advisors describe typical risk factors
- Students explain challenges in attendance or assignments

Problem Definition

- Objective: Identity at-risk students before midterm
- Scope: Attendance, assignments, quiz secores
- Success Criteria: At least 70% correct identification, improve

Which data should be collected for each business problem?

Machine Learning Pipleline

Aspect	Retail (Product Demand)	Healthcare (No-Shows)	Education (Student Support)
Data Collection	POS, inventory, promotions, seasonal/holiday data	Appointment logs, demographics, reminders, attendance history	Attendance, LMS activity, quiz/assignment scores
Relevance	Shows real buying patterns & seasonal effects	Past behavior predicts no-shows; reminders influence attendance	Engagement & performance indicate risk early
Quality Needs	Clean IDs, timestamps, no duplicates, fix missing values	Accurate statuses, anonymized data, consistent patient records	Consistent grading, clean logs, validated missing data
Quantity	12–24 months of sales; large transaction volume	6–12 months; thousands of appointments	4–6 weeks of early-term data for all students
Diversity	Multiple branches, weekdays/weekends, promo vs. normal days	Different departments, visit types, time slots	Various learning behaviors, assignment types, activity levels

Machine Learning Pipleline

Step 3: Data Cleaning and Preprocessing

Data Cleaning

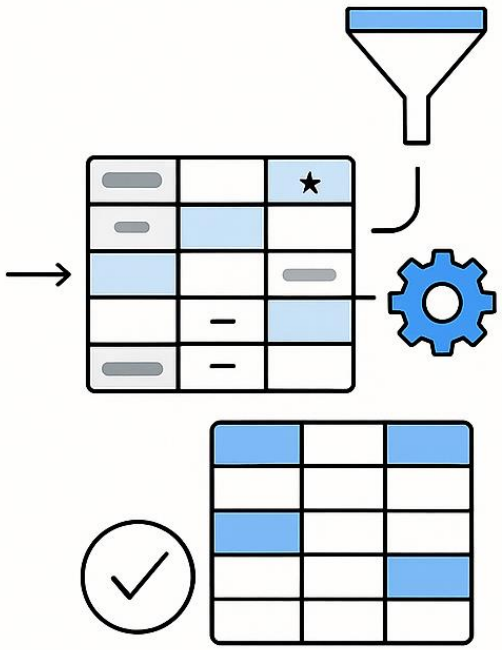
- Address issues such as missing values, outliers, and inconsistencies in the data

Data Preprocessing

- Standardize formats, scale values, and encode categorical variables

Data Quality

- Ensure that the data is well-organized and prepared for meaningful analysis



Step 4: Exploratory Data Analysis (EDA)

- Used to uncover hidden patterns and understand the dataset's structure
- Reveals trends and insights not easily visible at first glance
- Helps make informed decisions before modeling

Key Features of EDA

Exploration
Apply statistical summaries and visualizations to examine the data

Patterns & Trends
Detect underlying patterns, trends, anomalies, and challenges

Insights
Extract meaningful insights to guide further analysis

Decision Making
Supports feature engineering
Assists in selecting suitable rmodels

Machine Learning Pipleline

Step 5: Feature Engineering and Selection

Feature engineering and selection is a transformative process that involves selecting only relevant features to enhance model efficiency and prediction while reducing complexity.

Feature Engineering

Create new features or transform existing ones to capture better patterns and relationships

Feature Selection

Identify subset of features that most significantly impact the model's performance

Domain Expertise

Use domain knowledge to engineer features that contribute meaningfully for prediction

Optimization

Balance set of features for accuracy while minimizing complexity

STEP 6: MODEL SELECTION

For a good machine learning model, model selection is a very important part as we need to find a model that aligns with our defined problem, nature of the data, complexity of the problem, and the desired outcomes.

COMPLEXITY	DECISION FACTORS	EXPERIMENTATION
Consider the complexity of the problem and the nature of data when choosing a model	Evaluate factors like performance, interpretability and scalability when selecting a model	Experiment with different models to find the best fit for the problem.

Machine Learning Pipleline

Step 8: Model Evaluation and Tuning

Model evaluation involves rigorous testing against validation or test datasets to test accuracy of model on new unseen data. It provides insights into model's strengths and weaknesses. If the model fails to achieve desired performance levels we may need to tune model again and adjust its hyperparameters to enhance predictive accuracy.

EVALUATION METRICS
Use metrics like accuracy, precsisicin, recall and F1 score to evaluate model performance

STRENGTHS AND WEAKNESSES
Identify the strengths and weaknesses of the model through rigorous testing

ITERATIVE IMPROVEMENT
Initiate model tuning to adjust hyperparameters and enhance predictive accuracy

MODEL ROBUSTNESS
Iterative tuning to achieve desired levels of model robustness and reliability

Step 9: Model Deployment

Integrate with existing systems

Enable decision-making using predictions

Ensure deployment scalability and security

Provide APIs or pipelines for production use

Data Collection

What is Data Collection?

- Definition: Systematic process of gathering, measuring & analyzing information from relevant sources.
- Purpose: To obtain information to make informed decisions, analyze trends, and measure progress.
- **Key terms in Data Collection**
  - Data: Facts/figures to understand the problem.
  - Investigator: Person conducting the enquiry
  - Enumerator: Person collecting data under the investigator's direction.
  - Respondent: Person from whom data is collected.
  - Survey: Asking questions of people to describe characteristics.

Data Collection

Categorising Data Collection Methods

- Primary Data – collected first-hand for the specific research.
- Secondary Data – already existing, processed or published by others.
- Choice depends on research/project objectives, resources, and time constraints.

Primary Data

- Definition: Information gathered directly from original sources for the particular study.
- Advantage: tailored to research, high relevance, high control over quality.
- Considerations: Cost, time, resource intensiveness, potential biases.

Methods of Collecting Primary Data

- Interviews:
  - Direct personal investigation: Investigating directly at the source.
  - Indirect oral investigation: Via intermediaries (others who have information).

Data Collection

Methods of Collecting Primary Data

- Questionnaires:
  - Distributed via mail / online / face-to-face.
  - Two approaches: mailing with letter; enumerator delivers questionnaire.
- Observations:
  - Collecting data by watching behaviors/events as they occur naturally.
- Experiments
  - Manipulate one variable, compare outcomes (control vs experimental group).
- Focus Group
  - Small group discussions (6-12 participants) moderated to elicit attitudes/opinions.
- Information from Local Sources / Correspondents
  - Local contacts collect data across varied locations for the investigator.
  - Useful for wide geographic coverage; subject to reliability of correspondents.

Data Collection

Secondary Data

- Definition: Data that has already been collected, processed, and published by others.
- Advantages: readily available, low cost, time-saving.
- Limitations: may not exactly match research needs, issues with relevance, currency, data quality.
- **Method of Collecting Secondary Data**
  - Government publications (e.g., statistical abstracts, surveys).
  - Trade association data, research journals/papers.
  - International organizations (e.g., World Bank, ILO)
  - Data collected by organizations for internal use (business records, professors' research).
    - Potential value: unique data sets not publicly available.
    - Challenges: access, transparency, unknown quality/metadata.

Data Cleaning in ML

Data cleaning is a step-in ML which involves identifying and removing any missing, duplicate or irrelevant data.

- Raw data (log file, transactions, audio /video recordings, etc.) is often noisy, incomplete and inconsistent which can negatively impact the accuracy of model.
- The goal of data cleaning is to ensure that the data is accurate, consistent and free of errors.
- Clean datasets also important in EDA (Exploratory Data Analysis) which enhances the interpretability of data so that the right actions can be taken based on insights.

Data cleaning means fixing bad data in your data set.

Bad data could be:

- Empty cells
- Data in wrong format
- Wrong data
- Duplicates

Let's make an observation on a dataset.

titanic dataset.

Data Cleaning in ML

Why Data Collection?

Data Cleaning in ML

Navigating Common Data Quality Issues in Analysis and Interpretation

- **Missing values:** Gaps in data lead to incomplete insights, biased results, and unreliable conclusions.
- **Duplicate data:** Repeated entries distort statistics, inflate counts, and skew analysis outcomes.
- **Incorrect data types:** Mismatched formats (e.g., text instead of numbers) disrupt processing and introduce inaccuracies.
- **Outliers and anomalies:** Extremely high or low values can heavily influence statistical results and distort interpretations.
- **Inconsistent formats:** Variations in date formats, capitalization, or structure make data integration and cleaning more difficult.
- **Spelling and typographical errors:** Misspellings in text fields cause misclassification and incorrect grouping during analysis.

Data Cleaning in ML

Common Data Cleaning Tasks

1. Handling Missing Data

- Missing data is a common problem in datasets. Strategies to handle missing data include:
  - Removing Records: Deleting rows with missing values if they are relatively few and insignificant.
  - Imputing Values: Replacing missing values with estimated ones, such as the mean, median, or mode of the dataset.
  - Using Algorithms: Employing advanced techniques like regression or machine learning models to predict and fill in missing values.
- Removing Duplicates
- Standardizing Formats
- Dealing with outliers

let's with handling missing data implementation